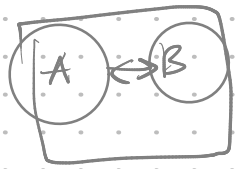
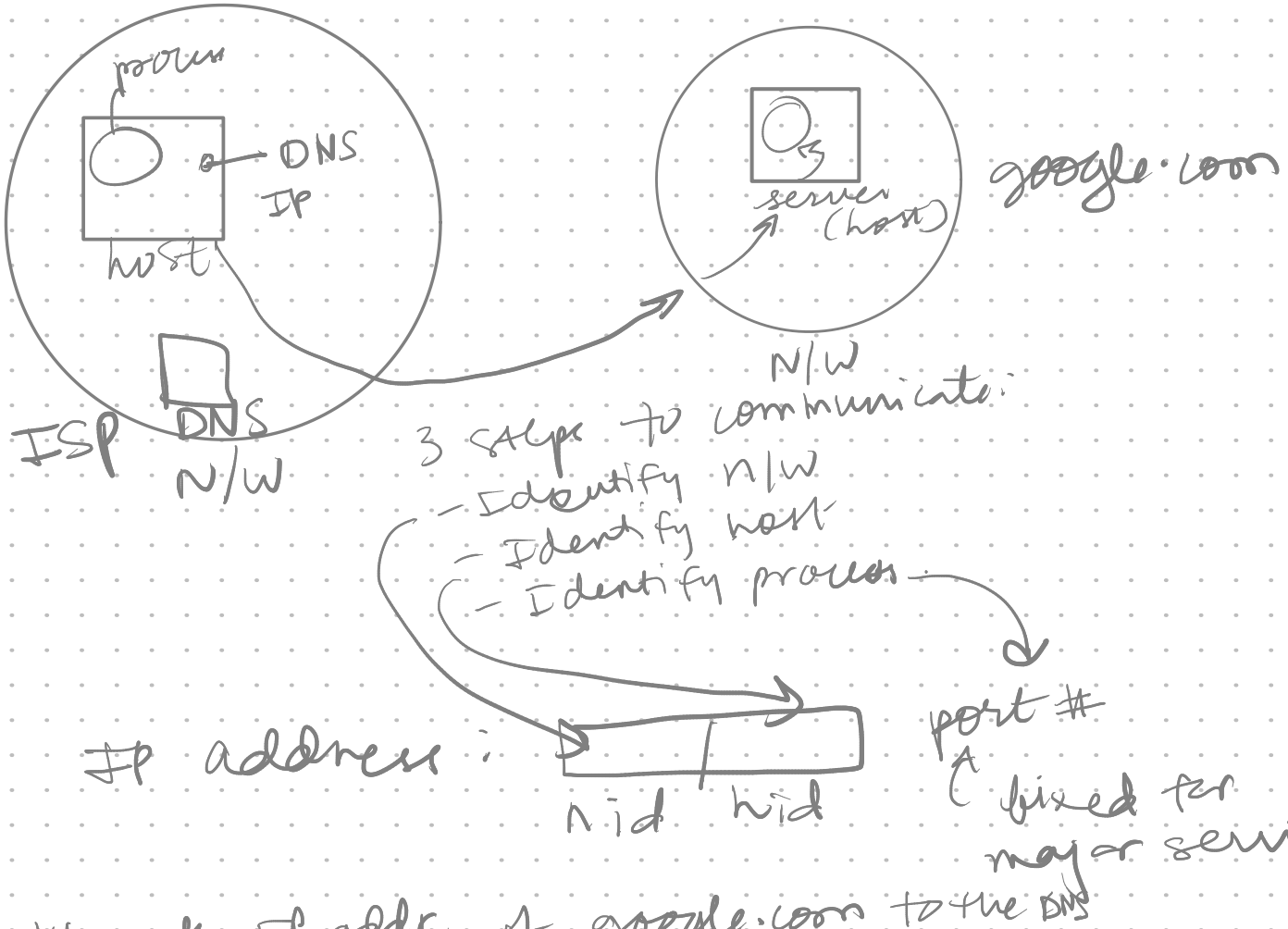
॥ श्रीगणेशाय नमः ॥

Computer Networks

Say there are 2 processes and they want to communicate. The OS will help them. (Inter process communication)



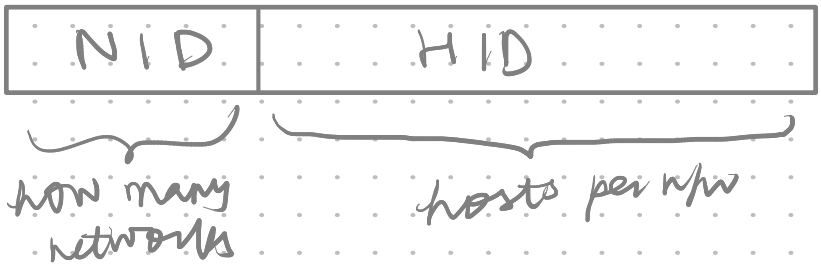
The issue is when there are 2 processes on different hosts which are on diff networks. CN is the H/W & S/W reqd.



we'll ask IP addr. of google.com to the DNS

In order to identify host, we also need a physical address. (MAC)

IP Address:



Initially (in ARPANET) 

problem: only $2^8 = 256$ n/ws are possible.

2^{24} IP addresses per n/w. too much.
→ 16 million.

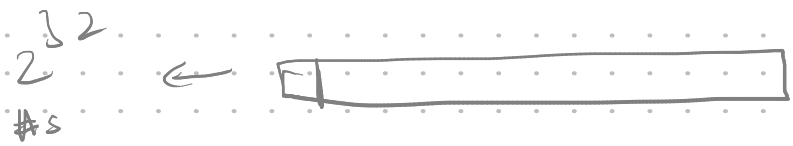
Say we have an n-bit # 

how many parts are we dividing the # space into?

n bits → 2^n #s

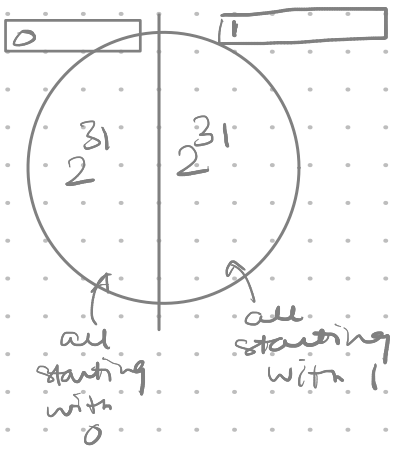
k → 2^k size of each = 2^{n-k}

2^k n/ws 2^{n-k} hosts.

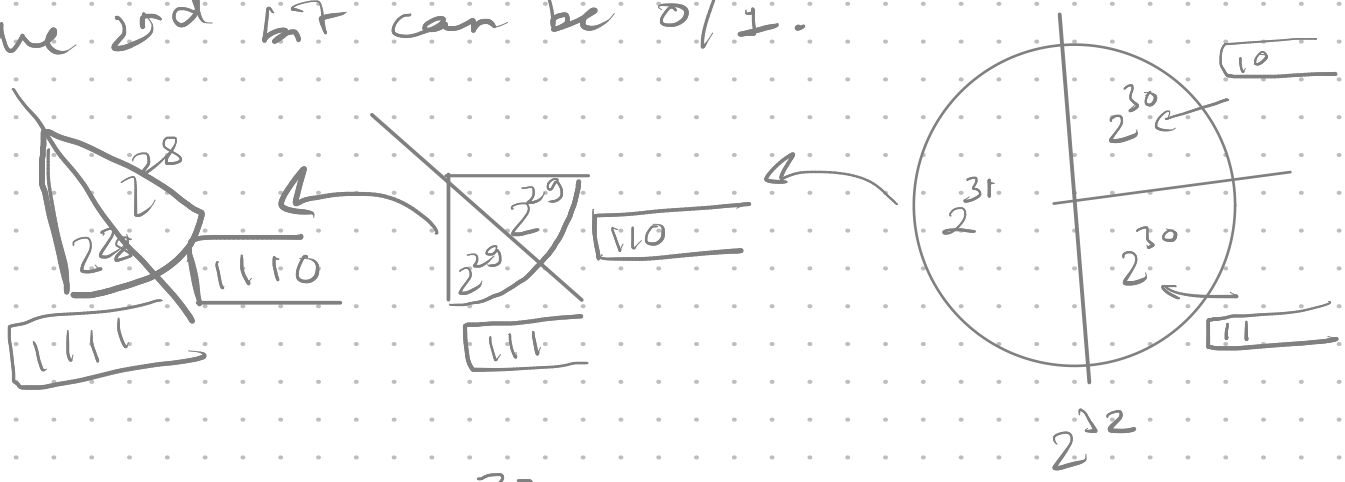


Think of it as a circle

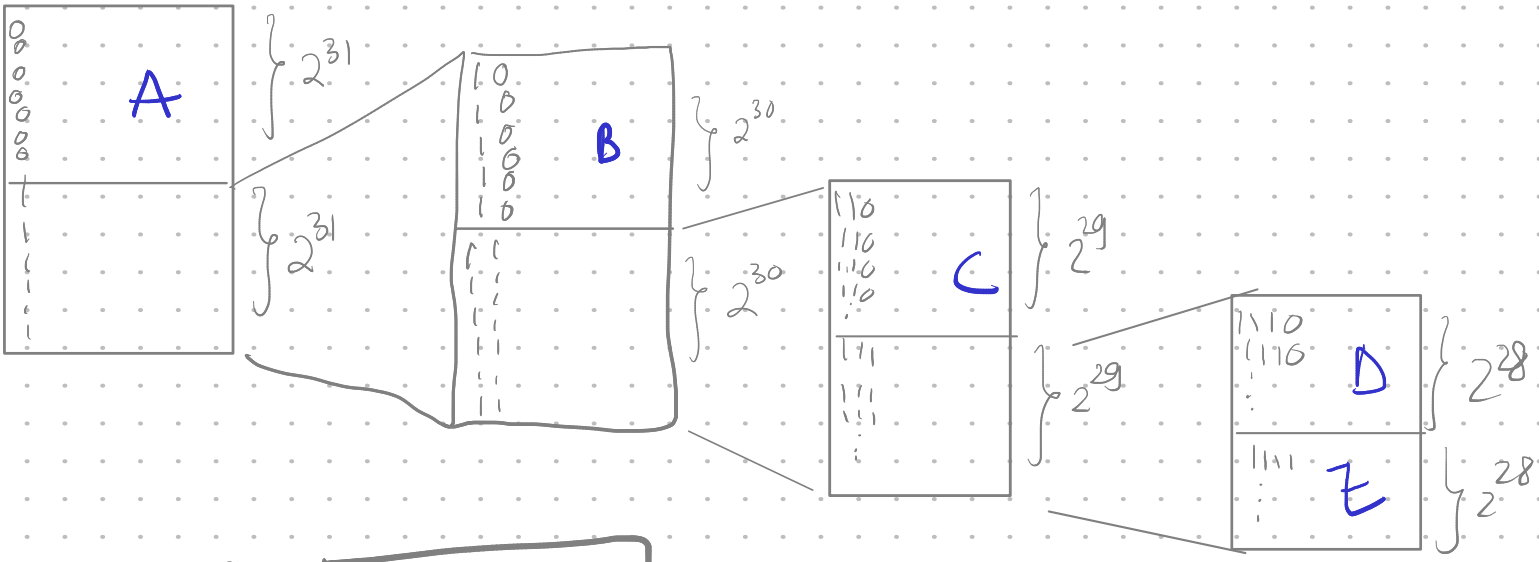
Say we choose the 1st bit.
0 or 1.



The 2nd bit can be 0/1.

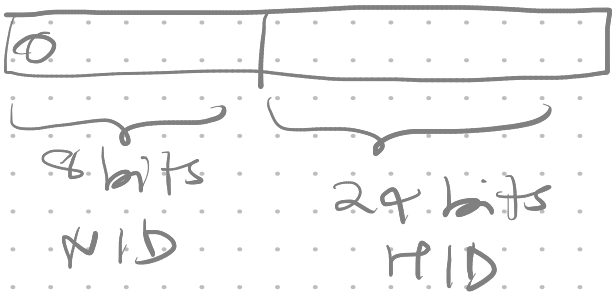


for 32 bits: 2^{32} IP addr. possible.



class A: 0

of IPs = $2^{31} = 2^{32-1}$



total # of n/w s in class A

$= 2^7 = 128.$

↑ in theory

32-bit IP is represented as dotted decimal #s.

n.n.n.n

↑ 8 bits as decimal

say

0000 1010 . 0000 0001 . 0000 0010 . 0000 0011

10 . 1 . 2 . 3

↳ 10.1.2.3 dotted decimal convention

---

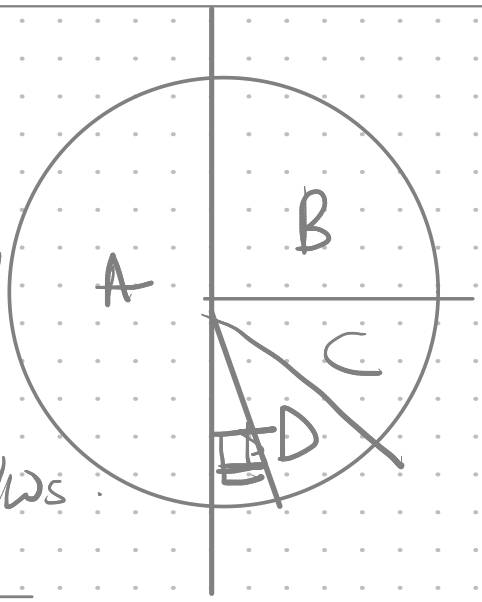
why divide into classes?

The way it is done with STD codes,

say [3 | 7] for big cities.

std device

My A will be for bigger n/ws.



Class A: [0 |] 8 bits — 2^{31} IP addresses

n/w

consider 1st 8 bits 0 _ _ _ _ _ _ _

range from 00000000 to 01111111

0 $(2^7 - 1) = 127$

1st octet byte from 0 to 127.

nn . nn . nn . nn

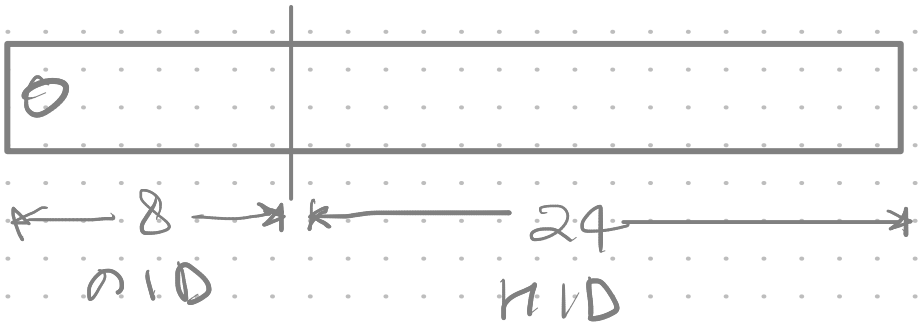
0 to 127

problem: we don't use 0 and 127 (for other purposes)

so 1 to 126.

range: 1 to 126 practically.

of n/ws in class A = 126.



total # of IP addr. in A = 2^{31}.

126 n/ws.

IP add./nw = 2^{24} = 16 m

of hosts/nw = $(2^{24} - 2)$

we have 16 million IPs but we can't use all of those:

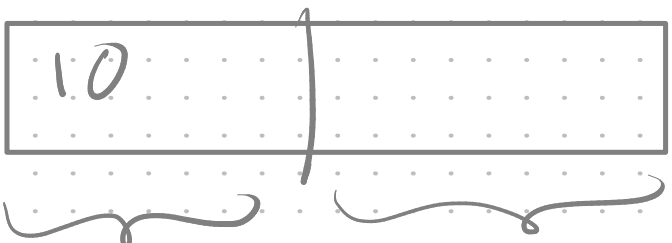
2 are reserved for special purposes.

So (# of hosts/nw) =

(# of IPs/nw − 2)

Class B: 10 — 30 bits

of IP addrs. in B = 2^{30}.



subtracting 2 not reqd.

of n/ws in class B = 2^{14}

size of each = 2^{16} = 64K.

↳ IP addr. in each n/w

of hosts in a n/w of class B

$= 2^{16} - 2$



10000000
to
10111111

i.e. 128 to 191

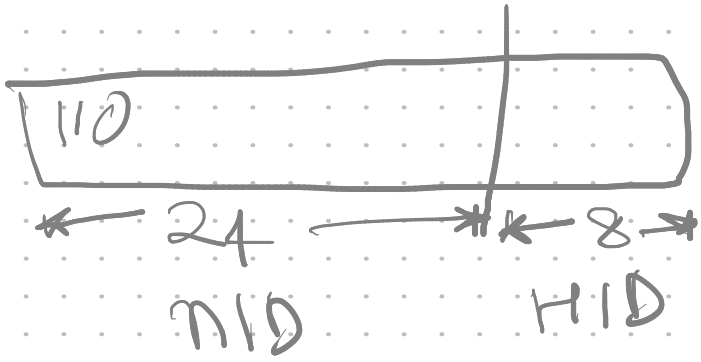
So

class A: 1 to 126

class B: 128 to 191

Class C: 110

of IP addr. in C $= 2^{29}$



of n/ws in C $= 2^{21}$

IPs in a n/w in C $= 2^8 = 256$

of hosts in a n/w in C $= 2^8 - 2 = 254$

Dotted Decimal:



11000000
to
11011111

i.e. 192 to 223

A: 1 to 126

B: 127 to 191

C: 192 to 223

e.g. Say 130.1.4.5

→ class B.

Size of a n/w in class A = 2^{24}

size of a n/w in B = 2^{16}

size of a n/w in C = 2^8

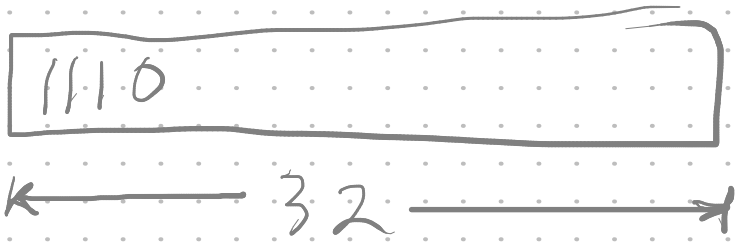
Do say we need 100 IP addr.

we'd buy a class C n/w.

1000 → class B

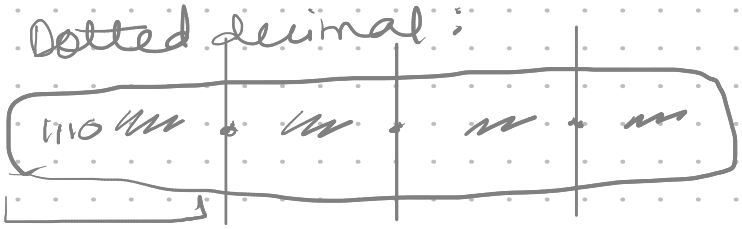
---

class D



of IP addr. in class D = 2^{28}

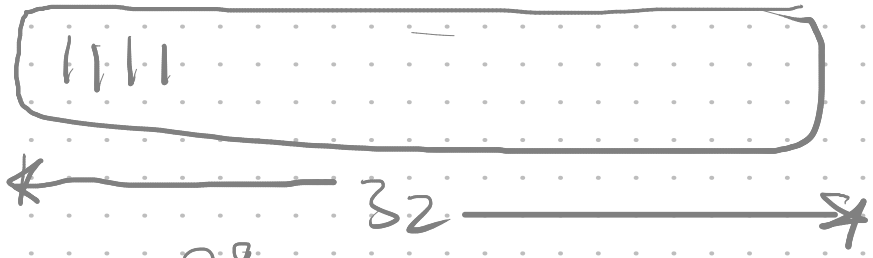
no NID and HID division.



11100000 to 11101111 i.e. 224 to 239

→ class D is reserved for multicasting

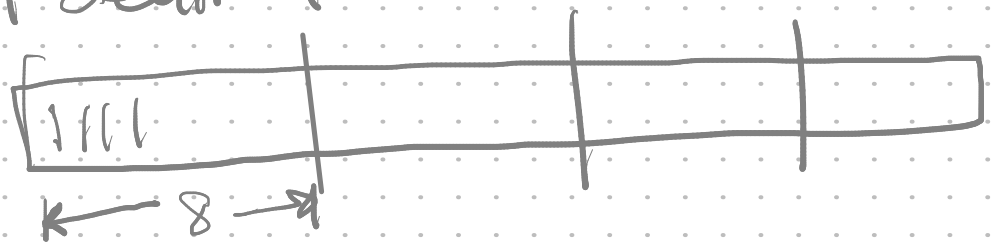
Class E:



of IP addr. = 2^{28}.

→ all are reserved.
(military purposes).

Dotted decimal:



11110000
to
11111111

i.e. 240 to 255.

A: 1 to 126

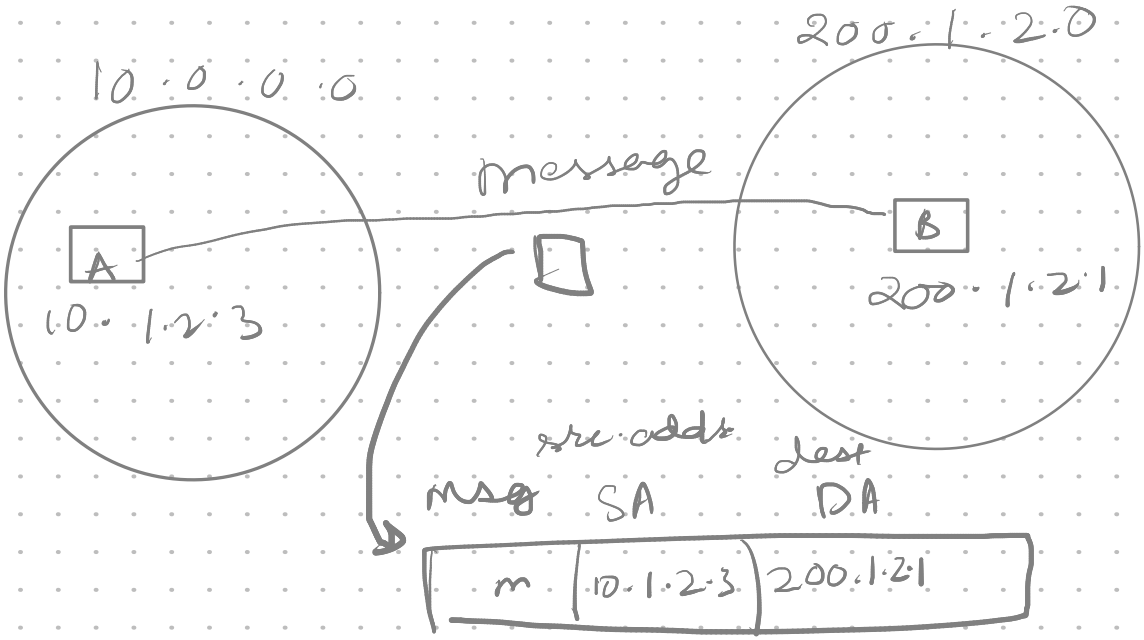
B: 128 to 191

C: 192 to 223

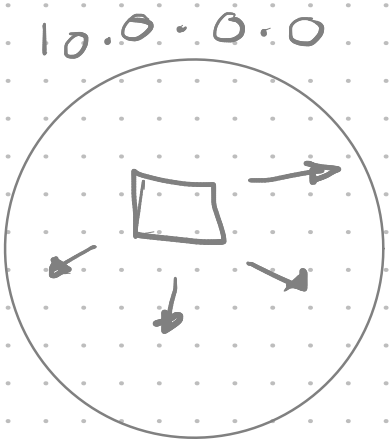
D: 224 to 239

E: 240 to 255

Unicasting : 1 → 1
host host



Limited Broadcasting:



Say the host 10.1.2.3 wants to send msg to everyone in the same n/w.

	S IP	D IP
msg	10.1.2.3	255.255.255.255

all 1s → limited broadcast address

10.1.2.3 → 10 . 1.2.3

(class A) NID HID

we'll call 10.0.0.0 (all 0s in HID) as the NID. (represents the entire N/w)

150.1.2.3

(class B)

150.1 . 2.3

NID HID

NID of entire N/w : 150.1.0.0

Ly, 200.1.2.3 (class C)

NID : 200.1.2.0 (all 0s in HID)

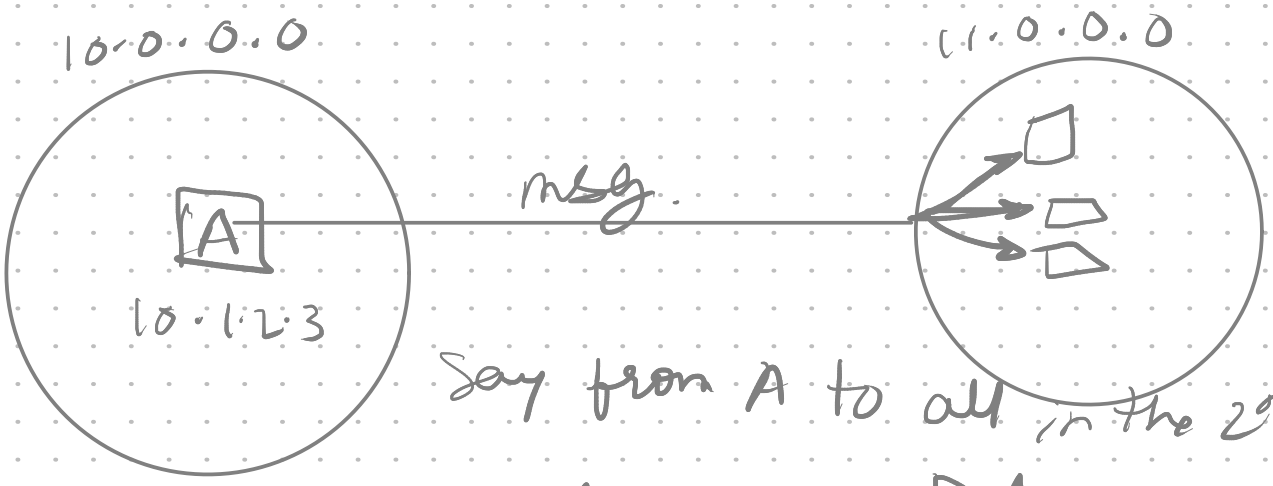
These addresses are special and cannot be assigned to any host.

HID replaced with zeroes.

So we can't use 00000000000000000000000000000000 all zeroes.

So we subtracted 1 from class A range

Another one:



Say from A to all in the 2nd n/w.

	SA	DA
msg	10.1.2.3	11.255.255.255

all 1s in HID.

Broadcast to all hosts in a diff n/w

↳ Directed Broadcast.

DBA
directed broadcast address.

Consider 1.2.3.4

NID: 1.0.0.0

DBA: 1.255.255.255

Ily for 130.1.2.3

NID: 130.1.0.0 (class B)

DBA: 130.1.255.255

For 200.1.10.100

NID: 200.1.10.0

DBA: 200.1.10.255

250.15.1.10

→ belongs to class E.

∴ there's no host / n/w or anything.

(valid NID).(0's in HID) → NID of n/w

(valid NID).(1's in HID) → DBA directed broadcast addr.

(all 1's).(all 1's in HID) → LBA limited

→ these 2 can't be given to hosts.

So -2 from # of hosts in class A.

From 1 host to 1 host : unicasting.

From 1 host to all within n/w :
limited broadcasting
(all 1's)

From 1 host to all in another n/w
direct broadcasting
(all 1s in HID)

So 2 addrs are wasted in any n/w.

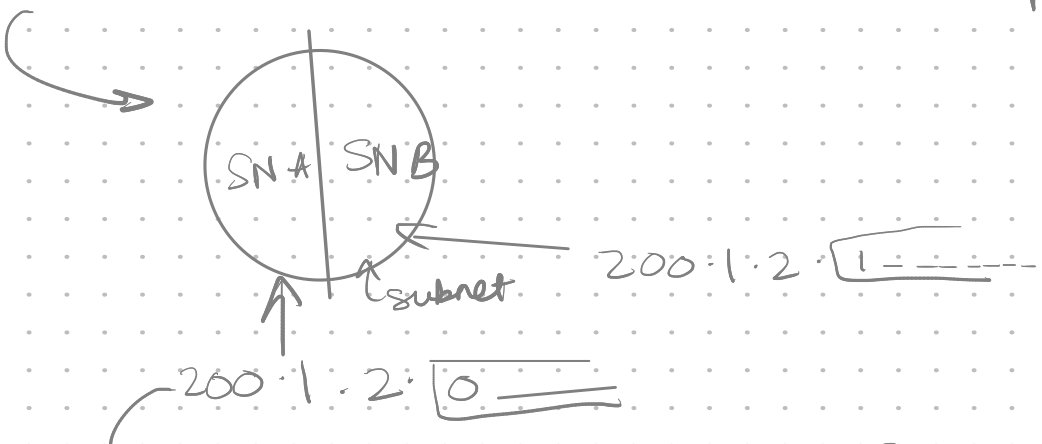
Multicasting:
1 host to many hosts in many diff n/ws

Subnetting

Say we have a NW 200.1.2.0 (class C)

200.1.2.0 → NID, 0 → HID

2^8 IP addresses possible



Range of SNA is from 200.1.2.0 _ _ _ _ _ _ _

0 0 0 0 0 0 0

to

1 1 1 1 1 1 1

i.e. 0 to 127.

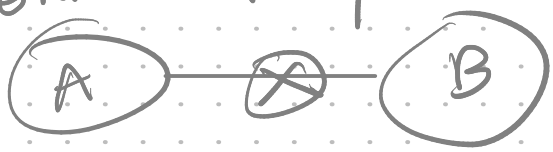
Range of SN B 200.1.2.10000000

to

11111111

i.e. 128 to 255

Now we have 2 NWs connected by a router.



NID or SID of A: 200.1.2.0] Host ID all zeros (first)

subnet ID

DBA of SA: 200.1.2.127] Host ID all ones (last)

NID or SID of SB: 200.1.2.128

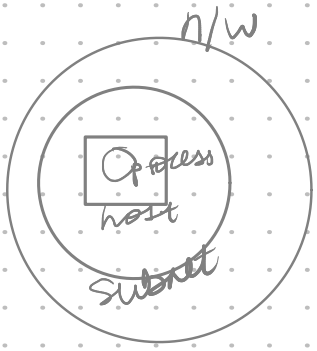
DBA of SB: 200.1.2.255

NID or SID or BID = 1st IP
n/w subnet block

DBA of any n/w = last IP

Why subnetting?
→ easy maintenance CA: 2^{24} IP addr. ≈ 16 M
→ security. (those n/w s that need to be protected could be placed in diff. subnets)
(access control rules can be applied)

Problem:



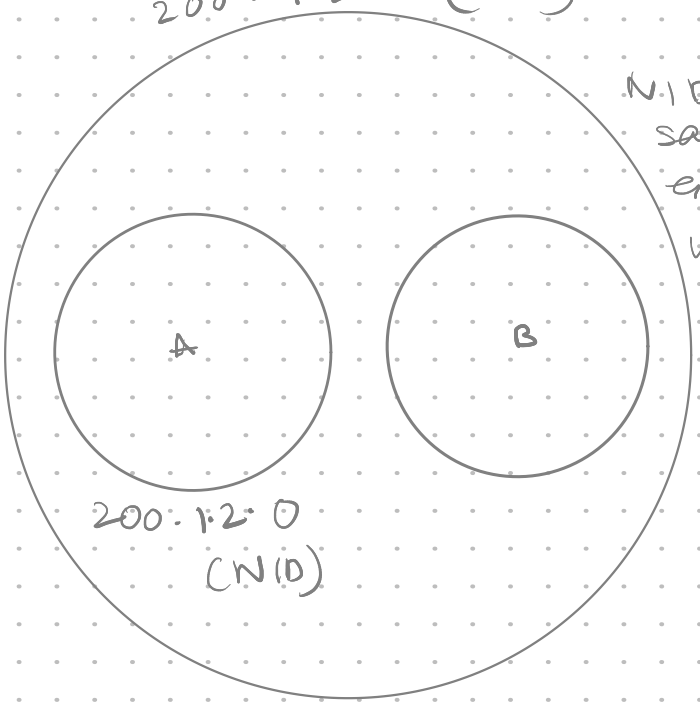
first n/w then subnet then host then process needs to be accessed.

Also, 200.1.2.0: # of hosts = $2^8 - 2 = 254$

H($2^7 - 2$) H($2^7 - 2$) Total = $2^8 - 4 = 252$

So there's a wastage of IP addresses.

200.1.2.0 (NID) ← for the entire n/w



NID becomes same for the entire n/w as well as the 1st subnet.

For any outsider, 200.1.2.0 will be treat as the entire n/w.

For anyone in the n/w, it'll be the SID.

Eg, for whole n/w,
DBA: 200.1.2.255

For subnet B,
DBA = 200.1.2.255

So if a packet has dest IP 200.1.2.255,
if packet is from elsewhere, it'll be given to all of the n/w.
If it is from the inside, say from subnet A, it'll be given to all of B.

Say a n/w is 130.1.0.0 (class B)

IP addr. → 2^{16} → S_1: 2^{15}, S_2: 2^{15}

130.1. _ _ _ _ _ _ _ _ . _ _ _ _ _ _ _ _

0 → NID 130.1.0.0 to DBA 130.1.127.255 (S_1)

1 → 130.1.128.0 NID to 130.1.255.255 DBA (S_2)

(130.1.10000000.00000000) to (130.1.11111111.11111111)

Q. 100.0.0.0 is NID. We want to divide into 2 subnets. range?

→ class A. 100.0.0.0
NID HID

Range

100._____._____._____

0 → 1

100.0.0.0
to
100.127.255.255

100.128.0.0
to
100.255.255.255

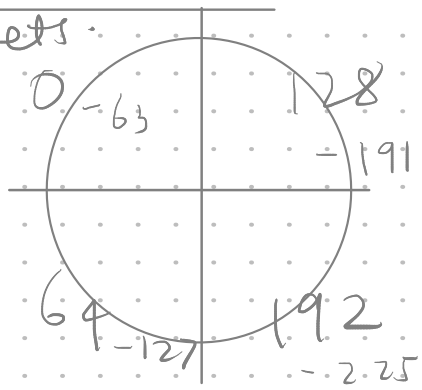
start: all zeroes
end: all ones

Q. 200.1.2.0 divide into 4 subnets.

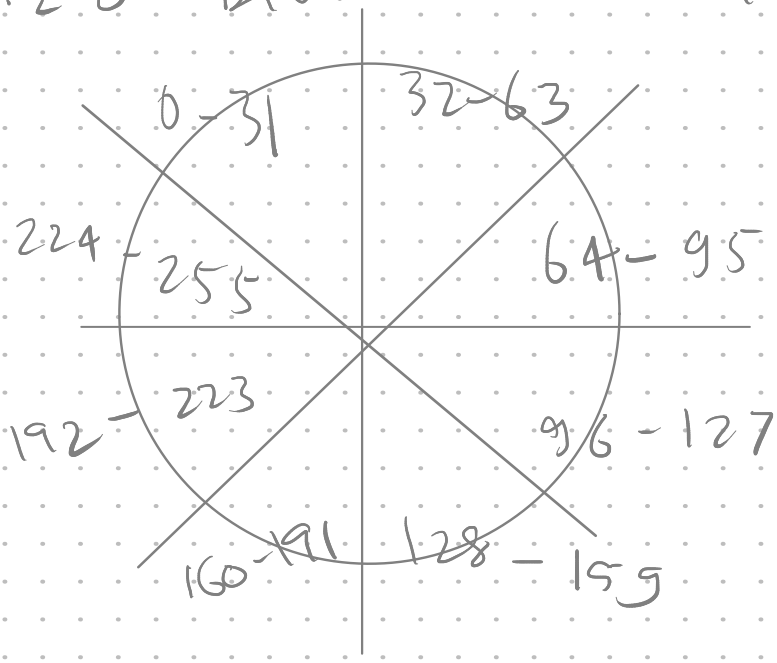
→ 200.1.2._____

2^7 2^6
128 64

can be 00, 01, 10, 11
↓ ↓ ↓ ↓
0 64 128 192



Q. 200.1.2.0 divide into 8 n/ws.

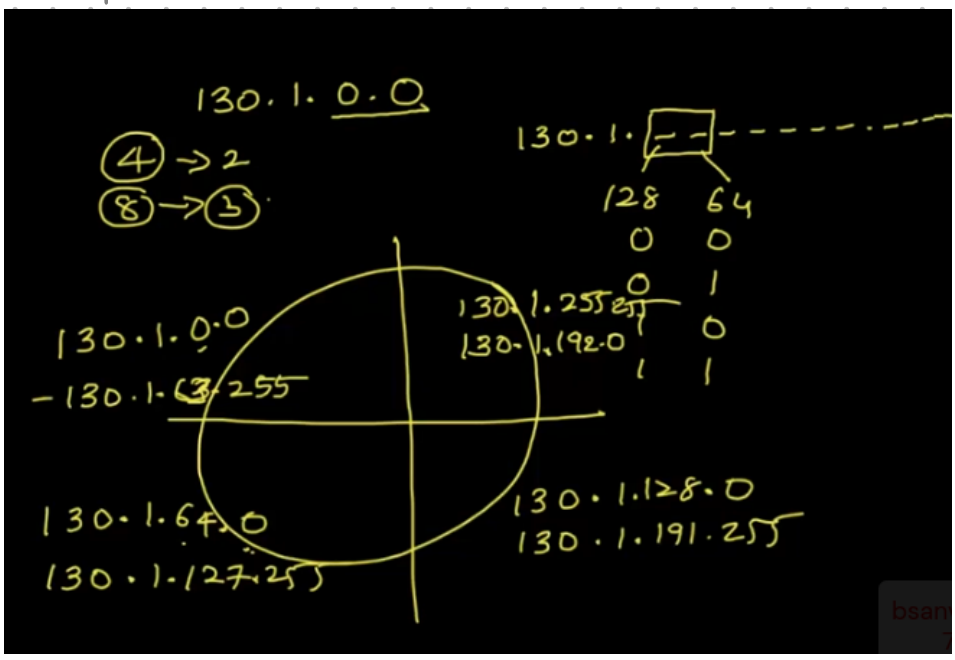


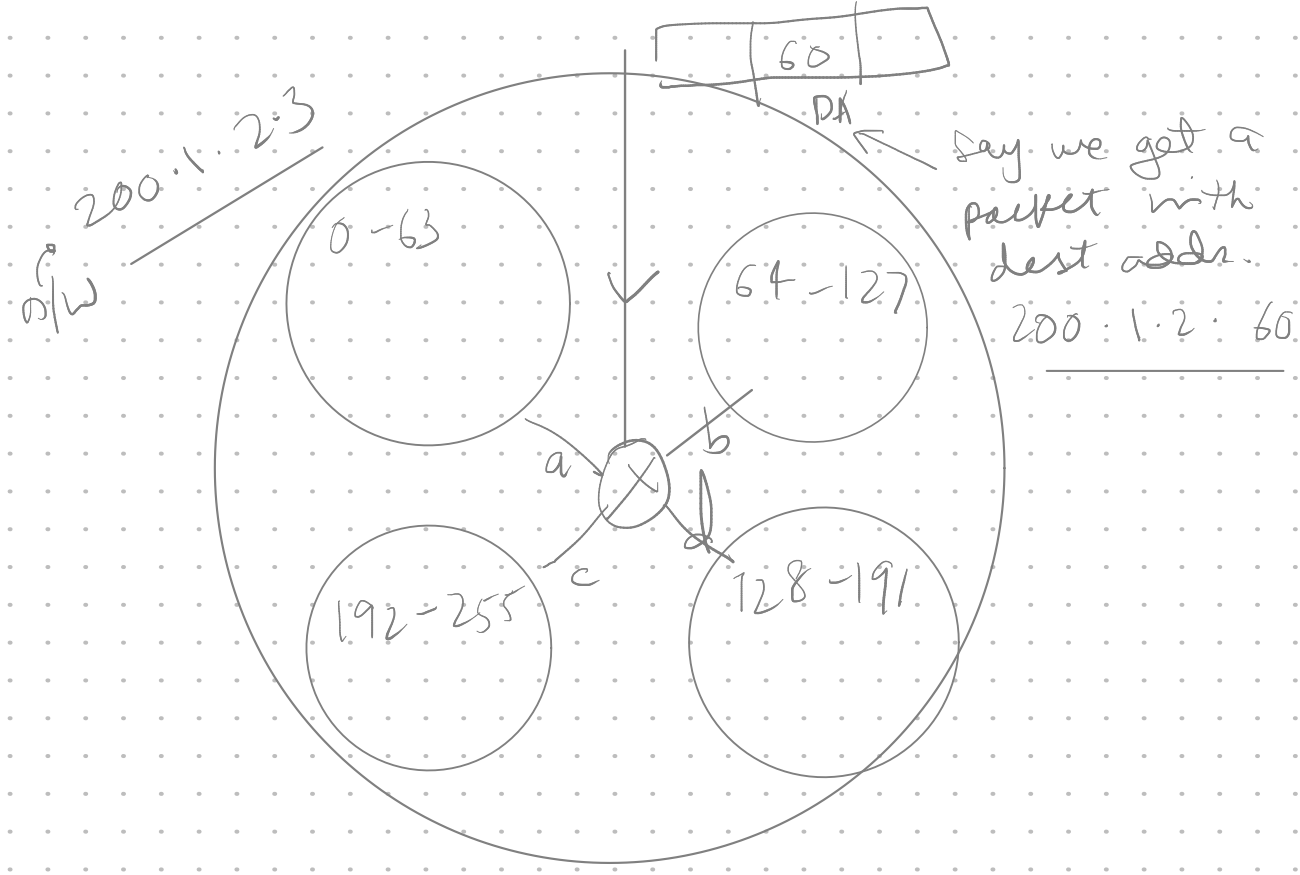
200.1.2. _ _ _ _ _ _ _ _

128 64 32

000 → 0
001 → 32
010 → 64
011 → 96
100 → 128
101 → 160
110 → 192
111 → 224

Another example:





Say we get a packet with dest addr. 200.1.2.60

What interface? → to send to

The router needs to know which interface to send to.

a, b, c, d.

There'll be some kind of code

if ip > 0 and < 63
then a

or sth. like that.

How do we do this without hard coding? (using explicit variables)

We will use subnet masks.

Subnet Mask

32 bit #

of 1's → NID + SID

of 0's → HID

class A: 8 bits → NID — No subnetting, SID = 0

24 bits → HID

Subnet mask:

11111111.00000000.00000000.00000000

or 255.0.0.0 — Default subnet mask for class B.

(255 ↑ NID, 0.0.0 ↑ HID)

Default SM for CB: 255.255.0.0

CC: 255.255.255.0

There's no subnet mask for class D or class E.

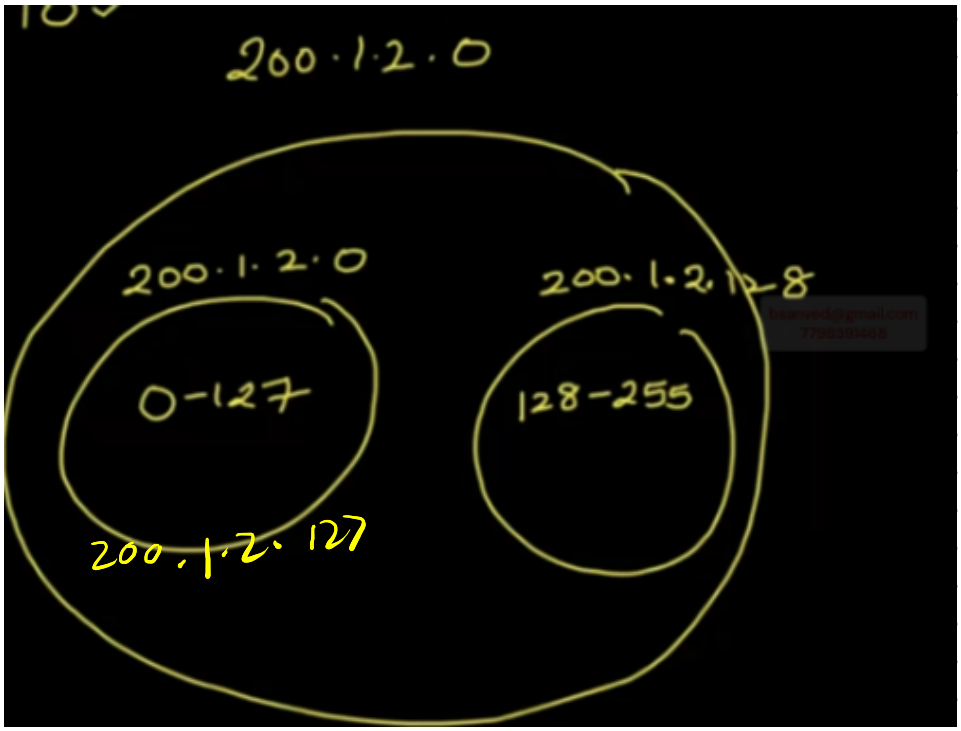
class C: 200.1.20 → 2 SNs.

255.255.255.128

(255.255.255 → NID; 128 → SID + HID)

For subnetting, we're borrowing bits from host ID.

200.1.2.0 is divided into 2 n/ws,



where to fwd a packet to 200.1.2.10?

SM for ~~both~~ = 255.255.255.128

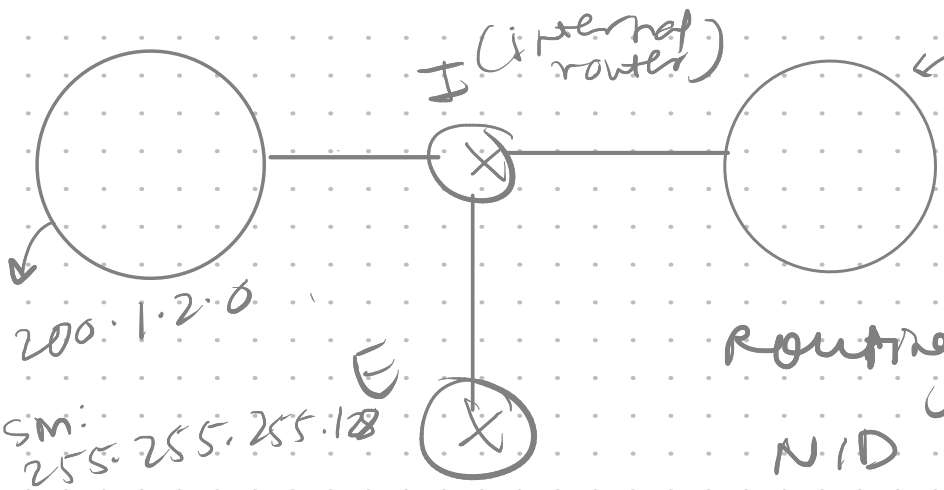
we do BITWISE AND ~~of both~~.

255 AND any # = that #

128 AND 10 →

```
  10000000
  00000010
  --------
  00000000   ← SID/NID
```

DO IF belongs to 200.1.2.0



Routing table at I

NID	SM	Interface
200.1.2.0	255.255.255.128	a_{eth0}
200.1.2.128	255.255.255.128	b_{eth1}
0.0.0.0	0.0.0.0	default

NOTE:
Same sized subnets will have same subnet mask

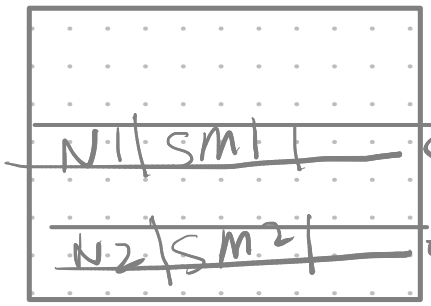
So I will take Dest IP (DIP) AND with SM.
If result same as NID then forward.
else go to 2nd entry.

If there aren't any matches with entries in the routing table we'll use the default entry that could go elsewhere.

BTW external router need not know about the internal subnetting.

NOTE:
If we put the default entry at the beginning, then all the packets will go to default.

Say



← match then

← match

we get 2 matching in the routing table

we'll compare # of 1's in SM_1 & SM_2

whichever is greater is picked.

As if # of 1's is greater then

VLSM: Variable Length Subnet masking